**THE UNITED REPUBLIC OF TANZANIA**
**PRESIDENT'S OFFICE**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**
**NJOMBE TOWN COUNCIL**

**FORM FOUR PRE- MOCK EXAMINATION APRIL -2024**

# 041 BASIC MATHEMATICS

**TIME: 3:00 HOURS** **16TH APRIL 2024**

INSTRUCTIONS

1. This paper consists of section A and B with a total of fourteen (14) questions
2. Answer all questions
3. Each question in section A caries six (6) marks while each question in section B carries ten (10) marks
4. All necessary working and answer for each question must be shown clearly
5. NECTA mathematical tables and non- programmable calculators may be used
6. All communication devices and any unauthorized material are not allowed in the examination room
7. Write your examination Number on every page of your answer booklet(s)

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## SECTION A (60 MARKS)
## Answer all questions

1. a) Six years ago, Aisha was $\frac{2}{3}$ times as old as Baedd. After four years, Baedd will be $\frac{4}{3}$ times as odd as Aisha. What is the age of Baedd.

   b) Solution A contains 65% of concentrated hydrochloric acid and solution B contains 90% of the same acid. How many litres of each solution must be used to make 200 litres 80% hydrochloric acid?

2. a) (i) Determine the value of $m$ from $\left(\frac{3}{4}\right)^{m} = \left(\frac{4}{3}\right)^{2m-m}$

   (ii) Simplify completely the expression, $\frac{4^{-3}\times 16^{3}}{8^{2}\times 32^{-1}}$

   b) Solve for $y$ from $2\log_3 y - \log_3 5 + 2\log_3 2 = 0$

3. a) A test on three subjects, Math, Biology and Kiswahili reveled the following:
   20 students passed Kiswahili
   5 students all the three subjects
   12 students passed Math and Biology and
   16 students passed Math and Kiswahili.
   Each students passed at least two subjects.
   (i) How many students passed Biology and Kiswahili?
   (ii) How many students did the test?

   b) In the joint sets $A$ and $B$, there 10 elements only in set $A$. If $\cap(A)$ is 5 more than $\cap(B)$; $\cap(A' \cap B')$ is 5 less than $\cap(A \cap B)$ and $\cap(A \cup B)$ is 3 less than $\cap(\mu)$. Find the probability that when an element is picked randomly from the universal set, $\mu$, will be the element from set $A$?

4. a) (i) On the $xy$ −plane, plot the points $A(2,6)$ and $(6,8)$. Measure the distance $\overline{AB}$ with your ruler. Compare the result when you use a formula for the distance between two points on a plane.
   (ii) Use your ruler to find the coordinates of M. where M is the midpoint of $\overline{AB}$ in (a)(i) above.

   b) If $\boldsymbol{a} = 2\boldsymbol{i} + 5\boldsymbol{j}$ and $\boldsymbol{b} = -4\boldsymbol{i} + 4\boldsymbol{j}$. Find the direction of the vector $\boldsymbol{c} = 2\boldsymbol{a} + \boldsymbol{b}$ using:
   (i) bearing (acute angles)
   (ii) bearing (three digit angle)
   (iii) direction cosine

5. The right angled triangle with sides of lengths $7x\ cm,\ 24x\ cm$ and $150\ cm$. If the length with $150\ cm$ is the hypotenuse side, calculate:
   (a) the perimeter of the triangle
   (b) Area of the triangle

6. a) In the preparation of Fanta orange drink, a bottle filling machine can fill $1,500$ bottles in 45 minutes. Determine the number of bottles the filling machine can fill in 4.5 hours.

   b) Find the value of $\boldsymbol{n}$ and $\boldsymbol{m}$ in the table below, if $x$ varies directly as $y$ and inversely as $w$

| $x$ | 8 | 6 | $\boldsymbol{m}$ |
|---|---|---|---|
| $y$ | 4 | $\boldsymbol{n}$ | 2 |
| $w$ | 2 | 3 | 4 |

7. a) The profit on an item costing Tsh 8000 is decreased by Tsh 480 when its sells price is reduced by 5%. Find:

   (i) The sells price.
   (ii) Percentage profit before reduction.

(b)The following list represents assets and liabilities of Mkinga's business as on 31st December, 2023

Creditors Tsh 100 000
Bank overdrdft Tsh 50 000
Stock Tsh 85 000
Premises Tsh 250 000
Debtors Tsh 150 000
Cash in hand Tsh 15 000
Furniture Tsh 42 000
Capital Tsh 392 000
From the information above use it to find:

(i) Total non-current assets.
(ii) Current ratio.
(iii) Acid test ratio.

8. a) The first, second and fifth terms of an arithmetic sequence are consecutive terms of a geometric sequence. Find the possible values of the common ratio.
   b) If 1, $x$, $y$ and 27 are in Geometric progression, find the sum of the first 10 terms.

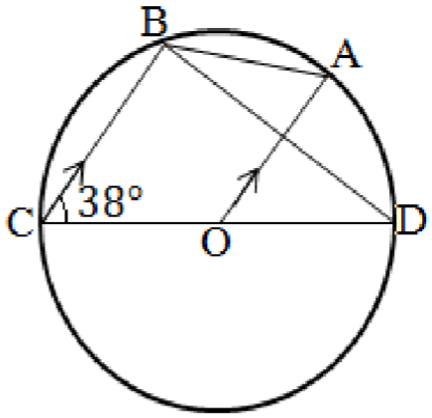
9. a) without using tables, find the value of sin 10 cos 20 + sin 20 cos 10

   b) To find the height of a tower a surveyor sets up his theodolite 100m from the base of the tower. He finds that the angle of elevation to the top of the tower is 30°. If the instrument is 1.5m above the ground. What is the height of the tower?

10. a) Years ago a man was 12 times as old as his son and 10 years from now a man will be twice as old as his son. Find their present age.
    b) Find the values of x that satisfy the equation log(x+5) + log(x + 2) = log4

**SECTION B (40 MARKS)**
**Answer all questions**

11. a) In the following figure, OA is the radius of the circle and parallel to BC, find the angle ABC if ∠OCB = 38°.



b) Prove that an angle inscribed in a semi-circle is $90^0$.

12. (a) VABCD is a pyramid whose base ABCD is a square with sides 6cm.The vertex v is vertically above N (the centre of the base ) and VN = 3cm.
Calculate.
(i) Length VA
(ii) Angle between the line VA and the plane ABCD. (iii)Volume of the pyramid.

(b) (i) A and B are two points on the equator with longitudes $40^0$ W and $25^0$E respectively. Taking the Earth to be sphere of radius 6370km, Calculate the shorter distance in km along the equator between A and B.

(ii) A ship sails from point $P_1$ (10◦S, 40◦E) to another port $P_2$ (10◦N , 40◦E) at 20 knots. If it leaves port $P_1$ at 12.00 mid night on Wednesday, when will it arrive at port $P_2$ ?

13. a) The line $2x - y = 1$ was shifted to another position by a vector $(-2,1)$.
Find its image at its new position.

b) A certain transformation $R$, maps the point P$(x, y)$ such $x' = x + 3y$ and $y' = 2x - 2y$
(i) Find the matrix $R$.
(ii) Determine the image of P(3,2) under this transformation.
(iii) Find $R^{-1}$

14. The function $f$ is defined as $f(x) = \begin{cases} x + 2 \ when \ x < -4 \\ 0 \ when -4 \leq x \leq 1 \\ 4 \ when \ x \geq 2 \end{cases}$ State domain and range of $f$.

(b) A firm wants to distribute parcels, using either a lorry or a van. A lorry can carry 200 parcels and costs Tsh 15,000 to hire, and a van can carry 80 parcels and costs Tsh 10,000 to hire. There only 20 men available, of whom only 15 are allowed to drive a lorry. The budget for the distribution is Tsh 260,000. Find the greatest number of parcels that can be distributed.